

Assignment 1: Taming the SQL Jungle

1. The Problem:

A SQL jungle is when there are too many messy and complex SQL queries in a system. It becomes hard to understand and manage the data. Two signs are: queries are very complicated, and different people write their own versions of the same query, causing confusion.

2. ETL vs ELT:

In ETL, data is cleaned before loading into the database. In ELT, data is loaded first and then cleaned later. The problem with ELT is that raw data can create confusion if not managed properly, leading to a SQL jungle.

3. Modularity:

Modularity means breaking big SQL queries into small parts called models. This makes the system easier to understand and fix. If there is a problem, we only need to fix one small part.

4. Engineering Discipline:

SQL should be treated like software code. We should use version control like Git and test queries regularly to avoid errors.

5. Anti-Pattern:

A common bad practice is copy-pasting SQL queries. This creates problems because if one query has an error, it gets repeated in many places.

Conclusion:

By organizing SQL properly and using good practices, we can make systems clean and easy to manage.